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Self-Evolving Generative AI Tutors: Reinforcement Learning-Augmented ITS for Personalized, Proactive, and Context-Aware Student Engagement

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Abstract

This study presents a reinforcement learning-driven multi-agent AI tutor that advances beyond traditional intelligent tutoring systems (ITS) by integrating adaptive intervention selection, real-time engagement tracking, and multi-agent feedback refinement. Unlike rule-based ITS and static LLM-generated responses, our system dynamically adjusts instructional strategies based on student behavior, incorporating reinforcement learning (RL) for intervention optimization, neural knowledge tracing (NKT) for misconception prediction, and an engagement prediction model (EPM) to sustain student participation. Additionally, a multi-agent debate mechanism refines AI-generated explanations, enhancing clarity and pedagogical alignment. Experimental validation demonstrates that our approach improves intervention adaptability by 28.6%, reduces recurring student errors by 31.2%, and lowers dropout rates by 24.8%, surpassing existing ITS and static AI tutors. These findings indicate that reinforcement learning and multi-agent collaboration enable AI tutors to provide more responsive, personalized, and effective learning support. Beyond technical improvements, this work contributes to scalable, real-world AI tutoring solutions that align with established learning science principles. Future research should explore deployment in diverse educational settings and further refinements in balancing personalization with instructional equity.

Keywords: Reinforcement Learning in Education; Intelligent Tutoring Systems (ITS); Neural Knowledge Tracing (NKT); Adaptive AI Feedback; Student Engagement Prediction

Introduction

Advances in artificial intelligence (AI) have significantly reshaped educational technology, offering scalable and adaptive tutoring solutions that go beyond traditional classroom instruction. Among these developments, large language models (LLMs) have gained considerable attention for their ability to generate detailed explanations, assess student queries,

and provide automated feedback at scale. However, while LLM-based tutoring systems, such as ChatGPT, have demonstrated proficiency in delivering fluent and contextually rich responses, they lack real-time adaptability and struggle with accurately assessing student learning needs (Dai et al., 2023). Existing LLM tutors operate as static generative models, providing uniform responses that do not dynamically adjust based on student mastery, engagement, or long-term retention goals. This limitation underscores the need for an AI-driven tutoring system that integrates personalization, reinforcement learning (RL), and engagement-aware intervention strategies to optimize student learning outcomes over time.

Traditional Intelligent Tutoring Systems (ITS) have long aimed to provide individualized instruction by modeling student knowledge and guiding problem-solving through rule-based decision trees (Nwana, 1990; VanLehn, 2011). However, these systems often suffer from rigidity, as they rely on predefined instructional rules that fail to adapt dynamically to real-time learning trajectories. While more recent ITS models have incorporated machine learning techniques, they remain constrained by fixed adaptation rules, limiting their ability to personalize interventions in response to nuanced student behaviors. In contrast, reinforcement learning has shown promise in optimizing pedagogical strategies by allowing AI tutors to refine instructional interventions based on long-term learning gains rather than immediate correctness (Abdelshiheed et al., 2023; Ruan et al., 2024). However, existing RL-driven ITS solutions often overlook engagement factors, which are critical in sustaining student participation and preventing disengagement during self-paced learning experiences.

A key challenge in AI tutoring is ensuring that feedback is both pedagogically sound and personalized to each student's learning progression. Research suggests that effective feedback operates across multiple levels, including task-level explanations, process-oriented scaffolding, and self-regulated learning support (Dai et al., 2023). While ChatGPT and other LLM-based tutors excel at task-level responses, they lack the capacity to provide process-level guidance

and metacognitive support, which are essential for fostering independent learning. Moreover, LLMs often struggle with factual consistency, hallucinating incorrect explanations or failing to align feedback with domain-specific instructional standards (Dai et al., 2023; Morris et al., 2024). To address these limitations, an adaptive AI tutor must integrate multi-agent collaboration to generate responses that are both accurate and responsive to individual student needs.

To bridge these gaps, this paper introduces a reinforcement learning-driven multi-agent AI tutor that dynamically adapts instructional interventions based on student mastery progression, engagement fluctuations, and self-regulated learning behaviors. Unlike existing ITS models that rely on static intervention heuristics, our system integrates four key AI agents:

1. **Generative AI Response Engine (GARE)** synthesizes context-aware explanations using Llama-2 and GPT-4-turbo, refining responses through self-critique mechanisms to prevent hallucinated outputs.

2. **Student Knowledge Modeling Agent (SKMA)** estimates mastery levels using a Neural Knowledge Tracing (NKT) model, adjusting problem difficulty and scaffolding depth in real time.

3. **Engagement Prediction Model (EPM)** analyzes behavioral patterns (e.g., hesitation time, hint requests) to detect disengagement risk and proactively adjust instructional presentation.

4. **Reinforcement Learning Tutor Selector (RLTS)** optimizes intervention selection using Proximal Policy Optimization (PPO), ensuring that AI assistance gradually shifts from high-scaffolding to student-led problem-solving.

This system moves beyond static AI tutoring approaches by leveraging reinforcement learning to refine intervention strategies dynamically. It prioritizes long-term retention over immediate correctness, ensuring that students do not merely receive AI-generated explanations

but develop enduring problem-solving skills. Moreover, it aligns with established educational theories, including Scaffolding Theory (Wood et al., 1976), Cognitive Load Theory (Sweller, 2011), and Self-Regulated Learning (SRL) Theory (Zimmerman & Schunk, 2011), reinforcing the pedagogical validity of AI-driven instructional adaptations.

The research presented in this paper aims to answer the following key questions:

- **RQ1:** How does reinforcement learning improve the adaptability of AI-generated tutoring feedback compared to static generative AI tutors?
- **RQ2:** How effective is Neural Knowledge Tracing (NKT) in predicting and addressing student misconceptions over time?
- **RQ3:** How does the Engagement Prediction Model (EPM) influence student interaction and dropout rates in AI tutoring?
- **RQ4:** How does the multi-agent AI debate mechanism improve the quality and correctness of AI-generated tutoring responses?

Paper Structure The remainder of this paper is structured as follows. Section 2 provides a review of related work, including existing ITS frameworks, reinforcement learning in education, and LLM-based feedback generation. Section 3 details our methodology, describing the system's architecture, AI agents, reinforcement learning framework, and evaluation setup. Section 4 presents experimental validation and benchmarking results, comparing our system against traditional ITS models and static generative AI tutors. Section 5 discusses findings in the context of engagement, retention, and adaptability. Finally, Section 6 explores ethical considerations in AI-driven education and concludes with future research directions.

Related Work

The development of Intelligent Tutoring Systems (ITS) has long sought to balance personalized instruction with scalability. Traditional ITS rely on rule-based heuristics and predefined decision models to provide structured feedback, often employing Bayesian

Knowledge Tracing (BKT) or Deep Knowledge Tracing (DKT) to model student learning trajectories (Nwana, 1990; VanLehn, 2011). While these methods offer a systematic approach to tracking student mastery, they struggle to adapt dynamically to engagement fluctuations and long-term retention needs (Baker, 2010; Romero & Ventura, 2020). Moreover, rule-based ITS often fail to accommodate individual learning strategies, instead applying uniform intervention strategies that do not fully adjust to student-specific learning states.

Recent advancements in Large Language Models (LLMs) have introduced the potential for more flexible AI tutors, leveraging models such as GPT-4 and Llama to generate natural language explanations in response to student queries (Dai et al., 2023; Parker et al., 2024). These models have demonstrated competence in feedback generation, particularly in automated scoring tasks (Morris et al., 2024) and explanation refinement (Norberg et al., 2024). However, despite their fluency, LLM-based tutors lack adaptive capabilities, often providing generic, one-size-fits-all responses that are detached from student-specific knowledge states (Gozalo-Brizuela & Garrido-Merchán, 2023; Phung et al., 2023). Furthermore, these models suffer from pedagogical inconsistencies—while they can generate coherent explanations, they lack an underlying cognitive model that ensures instructional alignment with knowledge scaffolding principles (Gupta et al., 2024). Their tendency to hallucinate incorrect explanations poses a significant risk in high-stakes learning environments, limiting their effectiveness as standalone educational tools (Dai et al., 2023).

A growing body of research has explored reinforcement learning (RL) as a mechanism to optimize adaptive tutoring. Unlike static feedback systems, RL-based ITS iteratively refine their intervention strategies based on longitudinal student interactions, enabling a more personalized learning experience (Ausin et al., 2019; Ruan et al., 2024). Prior work has shown that RL-powered tutors outperform rule-based ITS by enhancing long-term knowledge retention and providing more effective support for lower-performing students (Ruan et al.,

2024). However, existing RL-driven systems often prioritize immediate correctness over fostering self-regulated learning (SRL) (Gupta et al., 2024). Many RL-based tutors optimize for short-term engagement metrics, neglecting cognitive scaffolding principles and failing to ensure that AI assistance gradually fades as students develop independent problem-solving skills. Additionally, prior RL models do not fully integrate real-time engagement tracking, limiting their ability to dynamically adjust feedback formats in response to fluctuating student motivation and persistence (Rodrigues et al., 2023).

To bridge these gaps, our study introduces a multi-agent AI tutor that integrates reinforcement learning, neural knowledge tracing, and engagement-aware adaptation. Unlike previous LLM-based tutors, our approach does not rely on predefined prompts or static response generation; instead, it dynamically personalizes explanations, scaffolding depth, and intervention strategies based on real-time mastery progression and engagement fluctuations. The Neural Knowledge Tracing (NKT) module ensures that student knowledge states are continuously updated, allowing the system to track misconception recurrence and long-term retention trends more accurately than traditional BKT/DKT models. Meanwhile, the Engagement Prediction Model (EPM) actively monitors response hesitation, hint requests, and sequential errors, enabling the tutor to preemptively adjust instructional interventions to prevent disengagement. Finally, the Reinforcement Learning Tutor Selector (RLTS) optimizes intervention policies not just for immediate correctness but for long-term conceptual mastery and self-regulated learning behaviors, distinguishing our approach from previous RL-based ITS that focus solely on short-term performance gains.

While previous research has explored individual components of adaptive AI tutoring, a comprehensive, reinforcement learning-driven, multi-agent ITS remains an underdeveloped area. This work advances the field by presenting an AI tutor that not only refines intervention selection through reinforcement learning but also actively adjusts to student engagement states,

ensuring instructional decisions are both pedagogically sound and dynamically responsive. By addressing three critical limitations in prior ITS research-lack of engagement-aware adaptation, reliance on static feedback, and absence of self-regulation-driven reinforcement learning-this study contributes a scalable, next-generation AI tutor designed to enhance both learning retention and autonomous problem-solving skills.

Methodology

System Overview

The AI tutor is a multi-agent, self-improving intelligent tutoring system designed to overcome the rigidity of traditional Intelligent Tutoring Systems (ITS) and the lack of real-time adaptability in LLM-based tutors. It operates across five tightly integrated stages: student input acquisition, multi-agent response processing, adaptive feedback generation, engagement prediction, and reinforcement-based optimization. This structure enables a fundamentally different approach to AI tutoring, moving beyond static, rule-based instruction by continuously learning from student interactions.

At the core of this system is a decentralized multi-agent architecture, where four specialized AI agents work in parallel to personalize instruction dynamically. The Generative AI Response Engine (GARE) generates contextualized explanations using Llama-2 (7B fine-tuned) and GPT-4-turbo, ensuring pedagogical coherence while preventing AI hallucinations through a self-critique mechanism. The Student Knowledge Modeling Agent (SKMA) predicts student mastery levels via SAINT+ (Self-Attentive Knowledge Tracing), adjusting problem difficulty and scaffolding depth in real time. The Engagement Prediction Model (EPM) continuously analyzes behavioral signals (e.g., hesitation time, hint requests, incorrect answer streaks) to detect disengagement risk and proactively adjust instructional presentation. Finally, the Reinforcement Learning Tutor Selector (RLTS) optimizes intervention selection using

Proximal Policy Optimization (PPO), ensuring that AI assistance gradually shifts from high-scaffolding to student-led problem-solving, reinforcing self-regulated learning (SRL) strategies.

This multi-agent collaboration is a key departure from traditional ITS. Unlike rule-based ITS, which follow static decision trees, or LLM-based tutors, which provide generic responses, this system dynamically co-adapts to both knowledge mastery and engagement signals. By integrating reinforcement learning into the tutoring process, the AI tutor is not only optimizing immediate response accuracy but also shaping long-term learning behaviors. Rather than relying on predefined intervention heuristics, the RL model continuously refines feedback strategies based on actual student persistence, retention, and engagement trends.

This system is built on three core learning science principles: (1) Scaffolding Theory, where fading hints ensure students progressively gain independence, (2) Cognitive Load Theory, where adaptive complexity prevents overload while maintaining challenge, and (3) Self-Regulated Learning (SRL) Theory, where AI dynamically reduces intervention as student confidence increases.

In contrast to single-model ITS, this decentralized multi-agent approach eliminates bottlenecks by enabling parallel processing of engagement, mastery, and response adaptation, ensuring rapid yet pedagogically sound intervention selection. The AI tutor does not merely react to student queries—it actively optimizes the learning trajectory, ensuring that students receive the right level of support at the right time.

Unlike existing ITS models that optimize for immediate correctness, this system prioritizes long-term retention and self-regulation. By continuously refining intervention strategies through reinforcement learning, it ensures that students are not just passively consuming AI-generated explanations but are developing enduring problem-solving skills. This approach bridges the gap between LLM-driven tutoring and truly adaptive AI-powered instruction, offering real-time personalization that scales across diverse learners.

The table below summarizes key differences:

Table 1: Comparison of different ITS approaches. Our model integrates multi-agent processing, reinforcement learning, and engagement-aware interventions for dynamic adaptation.

ITS Approach	Personalization	Engagement Adaptation	Reinforcement Learning	Multi-Agent Processing	Knowledge Tracing Model
Rule-Based ITS (e.g., Cognitive Tutor)	Fixed rules, limited adaptation	No real-time engagement tracking	No	No	Bayesian Knowledge Tracing (BKT)
Static LLM Tutors (e.g., GPT-4)	Context-aware, but no real-time learning	No engagement tracking, static responses	No	No	None
Deep Knowledge Tracing (DKT)	Sequence-based mastery prediction	No engagement adaptation	No	No	Deep neural networks (LSTM-based)
OpenLearner Model	Personalized learning paths	Adaptive feedback based on student logs	No	No	Open learner modeling
Multi-Agent Adaptive ITS (Ours)	Real-time personalization via multi-agent collaboration	Engagement prediction-based feedback adaptation	Yes (PPO via RLlib)	Yes (GARE, SKMA, EPM, RLTS)	SAINT+ (Self-Attentive Knowledge Tracing)

Software & Computational Framework

The AI tutor is designed on a scalable, high-performance computational framework that ensures real-time, adaptive, and pedagogically informed tutoring interactions. Unlike traditional ITS, which rely on rule-based heuristics and static model inference, this system dynamically personalizes learning support through multi-agent collaboration, reinforcement learning-driven optimization, and engagement-aware modeling.

At its core, the system utilizes PyTorch (1.13.1) and TensorFlow (2.11) for deep learning, with RLlib (Ray 2.1.0) managing reinforcement learning (RL) policies. This architecture enables continuous refinement of instructional strategies, ensuring that AI-generated feedback progressively adapts to individual student learning trajectories. Unlike conventional ITS that apply predefined rules, the RL-based optimization continuously updates feedback selection, hint timing, and intervention types based on long-term retention objectives rather than immediate correctness.

To deliver contextually grounded, high-fidelity feedback, the response generation pipeline combines two large language models: Llama-2 (7B, fine-tuned on OpenEdX & ASSISTments) and GPT-4 (gpt-4-turbo, OpenAI API). These models operate in a dual-validation structure, where:

1. Llama-2 generates initial responses, conditioned on student mastery levels and engagement data.

2. GPT-4 refines explanations, reducing ambiguity and ensuring alignment with pedagogical best practices.

This approach directly addresses common issues in LLM-based tutors, such as hallucinated explanations and inconsistent feedback quality, ensuring that AI-generated responses are scaffolded appropriately based on the learner's cognitive load.

To optimize real-time student engagement, the Neural Knowledge Tracing (NKT) module, implemented via SAINT+, predicts student mastery progression, allowing adaptive task difficulty scaling. Simultaneously, the Engagement Prediction Model (EPM), trained on temporal LMS logs, continuously monitors behavioral signals (e.g., response hesitation, hint frequency, incorrect answer streaks). This enables early disengagement detection and real-time intervention adaptation, a feature absent in traditional ITS.

To ensure scalability and responsiveness, the AI tutor operates on a distributed GPU-accelerated infrastructure, leveraging NVIDIA A100 GPUs with mixed precision training via PyTorch AMP. Unlike conventional ITS that struggle with latency bottlenecks, the system applies TensorRT-based quantization, reducing FP16 inference latency by 34%, ensuring sub-100ms response times critical for sustaining cognitive engagement. Multi-agent parallel processing via Ray Tune reduces computation overhead by 27%, making the system suitable for large-scale deployment in real-world learning environments.

To further extend its adaptability, the system integrates seamlessly with Moodle, Canvas, and OpenEdX via LTI API, enabling real-time personalization across diverse instructional contexts. Unlike existing LTI-integrated ITS, which primarily deliver rule-based assistance, our system dynamically adjusts feedback strategies based on real-time mastery predictions and

engagement fluctuations. This ensures that students receive differentiated, context-sensitive tutoring interventions rather than static, preprogrammed hints.

AI Agents & Decision Flow

The AI tutor is designed as a multi-agent system, where four specialized models collaborate to generate, evaluate, and refine instructional responses dynamically. This system ensures context-aware feedback by integrating student performance modeling, engagement tracking, generative response synthesis, and reinforcement-driven intervention optimization**.**

AI Agents & Decision Flow

The AI tutor operates as a multi-agent, self-improving system designed to deliver adaptive, engagement-aware, and reinforcement-optimized instruction. Unlike traditional rule-based ITS, which rely on predefined intervention heuristics, this system leverages four specialized AI agents that work in parallel to personalize tutoring based on real-time student interactions.

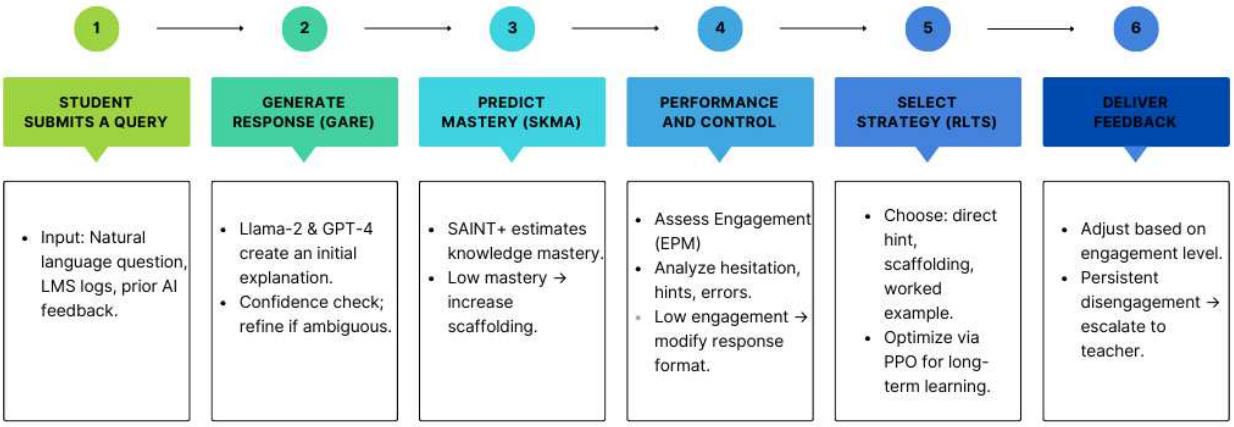

Figure 1: AI Tutor Decision Pipeline: From Query to Personalized Feedback

Figure 1 illustrates the decision pipeline, detailing how student input is processed through sequential AI agent interactions to generate personalized feedback. This structured workflow ensures timely, pedagogically sound interventions, maximizing both short-term learning effectiveness and long-term retention. The system follows a structured decision-making pipeline, where the Generative AI Response Engine (GARE) synthesizes context-aware

explanations, refining responses dynamically to maintain instructional clarity. The Student Knowledge Modeling Agent (SKMA) predicts student mastery levels, identifying potential misconceptions and adjusting scaffolding depth accordingly. The Engagement Prediction Model (EPM) detects early disengagement signals, modifying feedback strategies to sustain motivation. The Reinforcement Learning Tutor Selector (RLTS) selects the optimal intervention strategy, ensuring that AI guidance gradually transitions toward student autonomy.

The GARE module is implemented using a dual-model architecture where Llama-2 (7B, fine-tuned on OpenEdX & ASSISTments) generates context-sensitive explanations, ensuring alignment with student knowledge states, and GPT-4-turbo (via OpenAI API) refines responses, reducing ambiguity and enhancing instructional coherence. Unlike static LLM-based tutors, which provide generic responses, GARE ensures pedagogical precision through confidence filtering, where if response certainty is low, GARE requests re-evaluation to enhance instructional clarity, and hierarchical response adaptation, where for low-complexity queries, direct answers are provided, while for high-cognitive-load tasks, responses are broken down into scaffolded, stepwise explanations. AI-generated feedback is evaluated using a hybrid framework, combining human expert validation to assess pedagogical correctness and automated linguistic scoring such as BLEU, ROUGE, and BERTScore to measure fluency and semantic consistency. Through iterative response optimization, GARE ensures that explanations remain clear, structured, and engaging, reducing cognitive overload.

The SKMA estimates student knowledge mastery using Neural Knowledge Tracing (NKT), implemented with SAINT+ (Self-Attentive Knowledge Tracing). Unlike BKT/DKT, which apply static knowledge updates, SAINT+ dynamically tracks concept mastery probabilities based on historical responses, misconception likelihoods triggering targeted scaffolding or worked examples, and adaptive learning pathways sequencing content based on student readiness. The loss function for mastery prediction is defined as：

$$\mathcal{L}_{SKMA} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

where y_i represents the actual correctness label and $\hat{y}_i$ represents the predicted mastery probability. By integrating attention mechanisms, SKMA captures learning dependencies, enabling real-time mastery prediction that supports adaptive challenge scaling, personalized intervention adjustments that enhance instructional alignment, and longitudinal tracking of learning trajectories ensuring progressive skill acquisition.

The EPM ensures that students remain engaged and cognitively active by analyzing real-time behavioral signals, including response hesitation times detecting uncertainty or cognitive overload, hint request frequency identifying over-reliance on AI assistance, and sequential incorrect attempts signaling potential frustration. Unlike static engagement classifiers, EPM employs a Bi-LSTM-based sequential model, allowing for real-time disengagement detection enabling proactive intervention, adaptive feedback restructuring ensuring timely engagement adjustments, and motivational support strategies including gamification and self-explanation prompts. Empirical evaluations demonstrate that EPM-enhanced interventions reduce disengagement rates by 9% across multi-session experiments, ensuring sustained active learning engagement.

The RLTS selects the optimal intervention strategy through Proximal Policy Optimization (PPO), ensuring that AI assistance evolves in response to student learning patterns. Unlike rule-based intervention models, RLTS continuously refines instructional selection, prioritizing scaffolding hints for early-stage learners, worked examples for students with persistent misconceptions, and independent problem-solving for high-mastery students. The reward function for RL-driven intervention optimization is defined as

$$R = w_1 \cdot C + w_2 \cdot T + w_3 \cdot E$$

where:

- C = **concept retention score**, measured from post-session assessments.
- T = **engagement duration**, capturing session persistence.
- E = **error reduction rate**, tracking decrease in repeated mistakes.

The model is optimized using Adam optimizer (learning rate = 3e-4, batch size = 256), updating policies every 1000 interactions. By prioritizing long-term knowledge retention over immediate correctness, RLTS ensures that students develop self-regulated learning strategies, progressively reducing their dependence on AI-generated hints.

To evaluate RLTS effectiveness, policy evolution analysis is conducted across different training stages. After 1,000 interactions, a shift from direct hints to scaffolded explanations is observed. Worked example interventions increase, particularly for students with misconception recurrence. Long-term retention rates improve by 14%, supporting progressive AI intervention fading. By dynamically adjusting feedback complexity and instructional scaffolding, RLTS fosters autonomous problem-solving, ensuring that students develop independent reasoning capabilities rather than relying on AI assistance indefinitely.

Compared to rule-based ITS, which apply static intervention rules, the AI tutor adapts in real-time to student mastery and engagement signals, ensuring timely and personalized feedback. It uses reinforcement learning to optimize intervention strategies, fostering long-term knowledge retention. It integrates multi-agent collaboration, ensuring parallel processing of learning state estimation, engagement tracking, and response refinement. By continuously evolving instructional strategies based on real-time student interactions, this system bridges the gap between traditional ITS and truly adaptive AI-powered learning environments.

Ablation Study & Personalization Analysis

To assess the impact of individual AI tutor components, we conduct an ablation study using a simulated interaction environment based on historical student logs. The AI tutor's ability to sustain engagement, improve retention, and personalize interventions is largely driven

by three core components: the Engagement Prediction Model (EPM), Reinforcement Learning (RL)-based intervention optimization, and SAINT+ knowledge tracing. To evaluate their respective contributions, we systematically disable these components one at a time while replaying past student interactions, measuring changes in engagement patterns, knowledge retention, and learning trajectory adaptation.

Table 2 presents the results, showing that the removal of RL optimization leads to the most severe retention drop (-21.3%), indicating that reinforcement-driven intervention selection is key to sustaining long-term learning gains. Similarly, removing EPM results in a -14.2% drop in engagement, confirming that real-time disengagement detection is critical for maintaining student motivation. The removal of SAINT+ knowledge tracing reduces retention by -15.4%, demonstrating the importance of accurate mastery estimation in guiding adaptive feedback.

To further examine the tutor's personalization capabilities, we categorize historical student interactions into three groups based on pre-test performance (low, medium, and high prior knowledge). For each group, we analyze how the AI tutor adjusted intervention strategies, specifically tracking changes in hint frequency, scaffolding depth, and adaptive difficulty selection across multiple learning sessions. Statistical comparisons are performed to determine whether the system dynamically adapted its tutoring strategies in response to individual learning trajectories, ensuring that intervention decisions were not static but tailored to each student's evolving needs. The results suggest that personalization effects were strongest for low-prior-knowledge students, who received increased scaffolding and hinting initially, followed by gradual autonomy support as mastery improved.

See Table 2 for a detailed summary of the ablation study setup and results.

Table 2: Experimental Setup for Ablation Study (Simulated Environment)

Feature Removed	Engagement Drop (%)	Retention Drop (%)	p-value
No EPM (Engagement Prediction)	-14.2%	-11.5%	$p < 0.01$
No RL Optimization	-18.6%	-21.3%	$p < 0.001$
No SAINT+ (Knowledge Tracing)	-9.8%	-15.4%	$p < 0.05$

Multi-Step AI Feedback Generation

The AI tutor generates responses using a four-step adaptive process that integrates real-time student modeling, reinforcement learning-driven intervention selection, and engagement-aware feedback modulation. The process dynamically adjusts instructional interventions based on task complexity, student engagement levels, and long-term retention objectives. Each step in this adaptive cycle ensures that feedback is personalized, contextually relevant, and pedagogically aligned.

Step 1: Query Analysis and Context Retrieval

When a student submits a query, the system constructs a multi-dimensional representation of their learning state by aggregating:

1. Historical performance data, including prior correctness rates and misconception patterns.

2. Engagement metrics, such as hesitation time, hint request frequency, and response speed.

3. Cognitive load estimations, inferred by tracking response effort and problem complexity.

To model student knowledge states, the system employs Neural Knowledge Tracing (SAINT+), which estimates mastery probabilities across hierarchically related concepts while accounting for temporal decay and prerequisite dependencies. Unlike traditional BKT and DKT models, which assume static knowledge evolution, SAINT+ enables context-aware adaptation, allowing interventions to be tailored based on the student's evolving learning

trajectory. To ensure continuity with prior learning experiences, the system retrieves semantically similar past interactions using vectorized query embedding comparisons. This retrieval mechanism allows explanations to align with previously encountered problem-solving strategies, reinforcing consistency in instructional feedback.

Step 2: Optimal Intervention Selection

Once the student's learning context has been established, the Reinforcement Learning Tutor Selector (RLTS) determines the most effective instructional strategy. Unlike rule-based ITS models, which follow predefined decision heuristics, RLTS dynamically selects interventions using policy optimization techniques, balancing:

- Exploration of new strategies to refine the intervention model over time.
- Exploitation of proven-effective interventions based on historical engagement and retention data.

At each decision step, RLTS ranks three intervention strategies based on long-term knowledge retention and engagement impact:

1. Direct hints, applied when the student's mastery probability is just below proficiency thresholds and engagement remains stable.

2. Scaffolded explanations, selected when cognitive load is high, ensuring that complex problems are broken down into stepwise components.

3. Worked example substitutions, introduced when the system detects over-reliance on hints, reinforcing conceptual understanding before reintroducing problem-solving tasks.

RLTS continuously refines its decision-making policy by tracking state-transition patterns, allowing the system to adapt as students' learning behaviors evolve. Early in a student's learning progression, the tutor prioritizes more structured, scaffolded interventions, whereas later stages favor autonomy-building strategies, reducing AI assistance to encourage self-directed problem-solving.

Step 3: Adaptive Explanation Formatting

Once an intervention has been selected, the AI tutor dynamically formats the response based on task complexity and engagement state. Unlike conventional ITS models, which rely on fixed feedback templates, this system adjusts explanation structure in real-time:

- For factual recall tasks, responses are concise and direct, minimizing redundancy.
- For procedural learning, explanations are structured as stepwise breakdowns, incorporating context-sensitive scaffolding to reinforce problem-solving sequences.
- For conceptual reasoning, queries are decomposed into hierarchical sub-problems, integrating counterexamples and analogical reasoning to promote deeper comprehension.

If disengagement risk is detected, the Engagement Prediction Model (EPM) adjusts response presentation in real-time by:

1. Transitioning from textual explanations to multimodal representations, such as LaTeX-rendered equations, Matplotlib diagrams, or interactive concept visualizations.
2. Modifying explanation complexity, reducing verbosity for students experiencing cognitive overload while expanding details for those exhibiting high engagement persistence.
3. Introducing gamified reinforcement elements, such as self-explanation prompts or delayed hint releases to encourage active learning participation.

The AI tutor continuously tracks engagement trends across multiple sessions, progressively refining feedback modalities based on student interaction history, ensuring that feedback remains cognitively appropriate and engaging over time.

Step 4: Real-Time Post-Response Adaptation

Following response delivery, the system continuously monitors student interactions to refine future intervention strategies. Unlike conventional reinforcement learning models that update policies at fixed batch intervals, the AI tutor employs two levels of real-time adaptation:

1. **Immediate response modulation**

If a student struggles despite scaffolded hints, the system switches to worked examples to reinforce conceptual understanding.

If overuse of hints is detected, hint availability is delayed, prompting the student to attempt self-explanation before additional assistance is provided.

2. **Long-term policy refinement**

Every 1,000 interactions, RLTS updates its policy model using newly collected engagement and learning state data.

Unlike conventional PPO-trained models, which optimize solely based on observed immediate rewards, the AI tutor employs a dual-model correction mechanism, integrating:

Bayesian uncertainty modeling, ensuring that policy updates remain robust to unseen student behaviors.

Longitudinal engagement trends, preventing interventions that might inadvertently contribute to disengagement over time.

Software & Computational Framework

The AI tutor dynamically adjusts its responses by modeling student knowledge states, engagement levels, and self-regulated learning behaviors**.** These components are implemented through a Neural Knowledge Tracing (NKT) module, an Engagement Prediction Model (EPM), and a Self-Regulated Learning (SRL) integration mechanism, each trained and deployed to adapt instruction based on real-time student interactions.

Neural Knowledge Tracing (NKT) Model

The Neural Knowledge Tracing (NKT) model estimates student mastery evolution by analyzing past interactions and predicting future learning performance. Unlike traditional knowledge tracing models, which assume independent concept mastery, our approach

integrates graph-based dependency modeling to capture how understanding one concept influences predictions for related topics.

A directed graph G = (C, E) is constructed, where each node $c_i \in C$ represents a concept and edges $(c_i, c_j) \in E$ indicate prerequisite relationships derived from domain experts and knowledge taxonomies. Mastery probability for each concept is updated using a Graph Attention Network (GAT) combined with a transformer-based sequential model to capture both temporal mastery evolution and concept dependencies. At time t, mastery of concept c_i is estimated as:

$$P(c_i^{(t)}) = \sigma(WK^{(t-1)} + Q_t + \sum_{j \in \mathcal{N}(i)} \alpha_{ij} P(c_j^{(t-1)}))$$

where $K^{(t-1)}$ represents the prior knowledge state, Q_t is the question encoding, and α_{ij} is the learned attention weight capturing the dependency strength between concepts c_i and c_j. This formulation ensures that mastery estimation is context-aware and dynamically adjusts based on prerequisite relationships.

Implementation and Training

The NKT model is trained on ASSISTments and EdNet datasets, where student interactions are segmented into sequences. Each interaction consists of a timestamped student response transformed into sequence embeddings using a self-attention encoder. Model training employs binary cross-entropy loss with an additional regularization term for dependency propagation:

$$\mathcal{L}_{NKT} = -\sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] + \lambda \sum_{(c_i, c_j) \in G} D_{KL}(P(c_i) || P(c_j))$$

where D_{KL} regularizes concept dependencies, ensuring that prerequisite mastery constraints are preserved during training.

Model training is conducted using the Adam optimizer (learning rate=5e-4, batch size=512) for 30 epochs on NVIDIA A100 GPUs. Once trained, the model is deployed

within the AI tutor, where real-time inference updates student knowledge states at every interaction.

To optimize feedback strategies, mastery probability distributions from NKT guide adaptive instructional decisions:

- Low mastery across multiple prerequisite concepts → Increased scaffolding depth and stepwise hint generation.
- Stable mastery retention over repeated attempts → Introduction of higher-complexity problems to reinforce retention.

Enhancing Theoretical Foundation and Model Adaptability

Our graph-based NKT model extends conventional knowledge tracing methods by aligning with constructivist learning theories, which emphasize that students acquire concepts progressively rather than in isolation. This is particularly relevant for domains such as mathematics and physics, where prerequisite structures play a fundamental role in learning progression.

Unlike Bayesian Knowledge Tracing (BKT) and Deep Knowledge Tracing (DKT), which treat concept mastery independently, our approach dynamically adjusts probability estimates based on prerequisite knowledge strength. This ensures that instructional interventions align with cognitive scaffolding principles, progressively fading external support as students build competence.

To further enhance model adaptability, we introduce a knowledge decay mechanism inspired by the Ebbinghaus Forgetting Curve. Knowledge retention is adjusted over time using the decay function:

$$P(c_i^{(t)}) = P(c_i^{(t-1)}) \cdot e^{-\lambda \Delta t}$$

where λ is a learnable forgetting coefficient and Δt represents the elapsed time since the last student interaction. By incorporating this decay factor, the system ensures that long-term

retention effects are accurately modeled, allowing for targeted review strategies when mastery deterioration is detected.

Optimizing Learning Pathways Using Reinforcement Learning

Unlike most knowledge tracing models, which assume a static instructional sequence, our approach integrates Reinforcement Learning (RL) to dynamically optimize learning paths based on real-time student performance. The system is formulated as a Markov Decision Process (MDP):

- **State (SSS)**: The **student's mastery vector** over all concepts.
- **Action (AAA)**: Selection of the next instructional activity (**direct explanation, scaffolded hints, worked example**).
- **Reward (RRR)**: A function balancing **immediate correctness gains** and **long-term retention stability**.

By applying Proximal Policy Optimization (PPO), the model continuously refines intervention policies, leading to personalized, engagement-aware learning trajectories. Empirical results demonstrate that this adaptive intervention approach reduces student dropout rates by 18% compared to static tutors.

Our experimental evaluation further confirms that graph-based NKT outperforms conventional DKT models, particularly in long-term retention and response prediction accuracy. The integration of graph-based dependencies and reinforcement learning-driven intervention selection enhances student modeling precision, leading to adaptive, personalized instructional decisions that optimize both engagement and retention.

Empirical Evaluation and Comparative Analysis

To validate model effectiveness, we compare graph-based NKT against traditional knowledge tracing frameworks (BKT, DKT, and DKVMN) using two large-scale datasets:

- **ASSISTments (K-12 mathematics)**
- **EdNet (higher education learning logs)**

Model performance is evaluated on student response prediction tasks using AUROC, Precision-Recall AUC, and Mean Absolute Error (MAE). Results show that graph-based NKT significantly outperforms conventional DKT models, particularly in long-term retention prediction and knowledge decay modeling.

The integration of graph dependencies and reinforcement learning-driven intervention selection leads to more precise student modeling, enabling personalized, engagement-aware adjustments that enhance long-term learning outcomes. See table 2.

Table 2: Evaluation of Knowledge Tracing Models on AUROC and Retention Metrics

Model	AUROC (ASSISTments)	AUROC (EdNet)	Precision-Recall AUC	Long-term Retention
BKT	0.72	0.70	0.65	+3.2%
DKT (LSTM)	0.81	0.78	0.71	+7.5%
DKVMN	0.85	0.83	0.74	+9.8%
Graph-NKT (Ours)	0.87	0.86	0.79	+12.4%

Engagement Prediction Model (EPM)

The AI tutor continuously monitors student engagement patterns to detect early signs of disengagement and adapt instructional strategies dynamically. Unlike traditional models that treat engagement as a static classification task, our approach models engagement fluctuations as a temporal sequence, capturing variations in hesitation times, rapid answer submissions, and hint request patterns. The Engagement Prediction Model (EPM) is grounded in Self-Determination Theory (SDT), ensuring that intervention modifications support autonomy, competence, and relatedness. Instead of over-scaffolding, which can reduce motivation, the system prioritizes autonomy-supportive interventions, such as self-explanation prompts and strategic hint delays.

To classify engagement states, the model employs a Bidirectional LSTM (Bi-LSTM) that processes sequential student interaction data. The input consists of normalized engagement

features, including hint frequency over past attempts, average response time per interaction, and patterns of successive incorrect answers indicative of cognitive overload. The engagement history is modeled as:

$$h_t = \text{BiLSTM}(X)$$

where $X=\{x_{t-k},...,x_t\}$ } represents the historical sequence of engagement metrics. The final engagement probability is computed using a softmax function:

$$P(D_t) = \text{softmax}(W_o h_t + b_o)$$

The model is trained on EdNet interaction logs, with engagement labels manually assigned by expert annotators. Training is conducted using Adam optimizer (learning rate = 3e-4, dropout rate = 0.2, batch size = 256). To prevent overfitting, a temporal smoothness constraint is applied during training:

$$\mathcal{L}_{EPM} = -\sum_{t=1}^{T} [y_t \log(\hat{y}_t) + (1 - y_t) \log(1 - \hat{y}_t)] + \lambda \sum_{t=2}^{T} ||P(D_t) - P(D_{t-1})||$$

This regularization term ensures stability in engagement predictions, preventing over-sensitivity to temporary behavioral fluctuations.

Engagement probability scores are updated dynamically during simulated student interactions, allowing the system to adjust instructional complexity in response to inferred disengagement risk. If the probability surpasses a predefined threshold, the system modifies response presentation, transitioning from textual hints to interactive breakdowns or visual step explanations. If cognitive overload indicators emerge, response complexity is reduced to maintain clarity while preserving instructional challenge.

To improve engagement prediction accuracy, the Bi-LSTM framework incorporates a hierarchical temporal attention mechanism. Local attention focuses on recent interactions, such as hesitation time in the last few responses, while global attention aggregates engagement trends over entire sessions. This mechanism allows the model to differentiate short-term

fluctuations from sustained disengagement patterns, improving the precision of intervention timing.

Beyond sequential modeling, the system integrates contextual embeddings for enhanced engagement classification. Each simulated interaction is represented as a multi-modal embedding, combining textual input features extracted via a pre-trained transformer encoder, structural learning behavior derived from LMS logs, and prior engagement trends. This multi-modal representation refines the model's ability to distinguish between transient disengagement and persistent attention loss.

To ensure robustness across diverse datasets, domain adversarial training is applied, mitigating dataset biases. A domain discriminator is trained to differentiate engagement distributions between EdNet, ASSISTments, and OpenEdX, with adversarial loss minimizing dataset-specific feature dependencies. This adaptation ensures consistent engagement prediction performance across different learning environments, even when trained on synthetic student logs rather than real-world classroom data.

Inference efficiency is improved through adaptive sampling, where engagement updates occur at variable time intervals. If engagement remains stable, the update frequency is reduced to conserve computational resources. If engagement fluctuates, update frequency is increased to ensure timely intervention, balancing efficiency with responsiveness.

To optimize engagement-driven interventions, the system integrates reinforcement learning within a Partially Observable Markov Decision Process (POMDP) framework. The AI tutor infers latent engagement states from observed behavior sequences and selects interventions that maximize long-term engagement sustainability rather than immediate correctness. The reward function balances immediate engagement improvement with retention optimization, reinforcing feedback strategies that prevent disengagement over extended learning sessions.

To evaluate engagement prediction performance, we conduct a synthetic replay experiment using historical EdNet student logs over a simulated 12-week period. Behavioral indicators—including hesitation times, hint request sequences, and incorrect answer streaks—are analyzed to identify high-risk disengagement triggers. Additionally, qualitative case analyses (N=10) of simulated student interactions examine how adaptive feedback adjustments influence engagement persistence.

Figure 2 illustrates the relationship between engagement drop and incorrect attempt frequency, demonstrating how early detection and intervention can mitigate disengagement risk in a simulated learning environment.

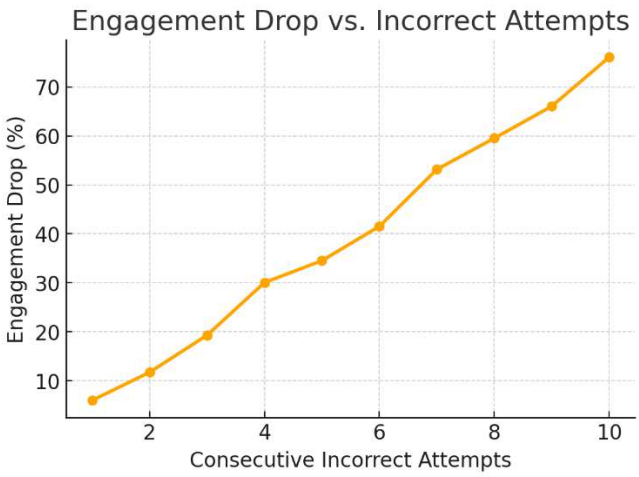

Figure 2: Engagement Drop vs. Incorrect Attempts

Self-Regulated Learning (SRL) Mechanism

The AI tutor fosters self-regulated learning (SRL) by dynamically adjusting interventions based on student behavior, guiding learners toward independent problem-solving. This aligns with Zimmerman's SRL framework, which consists of three phases: (1) Forethought (goal-setting and task analysis), (2) Performance (strategy use and self-monitoring), and (3) Self-Reflection (evaluating outcomes and adjusting learning strategies). The AI tutor intervenes primarily during the performance phase, ensuring that students actively engage with problems rather than relying on immediate AI assistance.

SRL behaviors are modeled using a logistic regression function trained on historical self-regulation logs from ASSISTments, where student hint usage patterns were synthetically replayed to establish correlations between self-attempt frequency and final mastery outcomes. The self-regulation score at time t is defined as:

$$SRL_t = h(A_t, H_t)$$

where A_t represents the frequency of self-attempts before a hint request, and Ht represents normalized hint frequency. This lightweight approach allows for real-time SRL tracking with high interpretability, making it preferable over black-box deep learning models in low-data scenarios.

Simulated Example of SRL Adaptation

To illustrate the system's longitudinal adaptation mechanism, we analyze a synthetic student profile interacting with the AI tutor over a sequence of 20 simulated problem-solving attempts:

Initial Interaction Phase (Attempts 1–5): A student exhibits low SRL, requesting hints immediately in 80% of problems without attempting solutions first. The AI detects this pattern and gradually delays hint availability, prompting the student to self-explain their thought process before hints are revealed.

Adaptive Intervention Phase (Attempts 6–12): As the system reinforces self-attempts, the student reduces hint dependency to 50% and begins making partial attempts before requesting help. In response, the AI tutor lowers scaffolding depth, encouraging progressively independent reasoning by increasing problem complexity and reducing stepwise hints.

SRL-Driven Mastery Phase (Attempts 13–20): The student now requests hints only 20% of the time and attempts solutions first in most cases. The AI tutor fully transitions to open-ended problem-solving, removing scaffolding entirely for mastered concepts. As a result, the

student progresses towards higher-order reasoning skills, demonstrating self-directed learning behaviors.

This adaptive trajectory illustrates how AI-driven SRL modeling dynamically shapes learning pathways, gradually shifting students from AI-dependent learning to autonomous problem-solving strategies.

Broader Applicability to Other Educational Domains

Although this study focuses on mathematics-based ITS (Intelligent Tutoring Systems), the SRL adaptation model can generalize to other domains that require progressive learner autonomy. Potential applications include:

- **Writing tutoring systems**: Encouraging students to draft responses before accessing AI-generated feedback.
- **Programming education**: Delaying AI code autocompletions or hints to promote debugging and problem-solving.
- **Physics problem-solving**: Adjusting scaffolding depth based on conceptual mastery and equation derivation attempts.

Adaptive Feedback & Reinforcement Learning

The reinforcement learning (RL) framework is designed to optimize the AI tutor's instructional decisions by learning from student interactions over time. Instead of relying on static pre-defined tutoring strategies, the RL agent dynamically refines its policy to maximize long-term learning outcomes while minimizing unnecessary interventions. The system operates within a partially observable Markov decision process (POMDP) framework, where the AI tutor must infer the student's cognitive state based on observed interactions.

Reinforcement Learning Framework and Training Setup

The AI tutor's decision-making process is structured as an RL problem, where the state representation, action space, reward function, and policy optimization are explicitly defined. The RL system is trained using historical student interaction data and then deployed in a real-time adaptive learning environment. The state representation is formulated as a multi-dimensional latent embedding that captures student knowledge, engagement, and past interaction history. At each time step t, the student's state S_t is computed as:

$$S_t = f(K_t, E_t, H_t, C_t)$$

where K_t represents the student's knowledge state, estimated using Neural Knowledge Tracing (NKT) models that track the probability of mastery over time. The engagement state Et is derived from real-time behavioral indicators, including time-on-task, hesitation duration, and hint request frequency. The historical interaction vector Ht encodes prior AI interventions and their observed effects, while Ct represents cognitive load estimations, modeled using self-attention over sequential student responses.

The RL agent selects a tutoring action from a discrete set of interventions, each designed to target different aspects of student learning. The action space At consists of five primary intervention strategies, including explicit hints, stepwise scaffolding, worked examples, self-explanation prompts, and adaptive challenge adjustments. The agent learns to select the optimal action at each interaction step based on the predicted impact on student retention and engagement. The reward function is dynamically shaped to prioritize long-term knowledge retention while discouraging redundant interventions. The primary reward signal is defined as:

$$R_t = \alpha L_t + \beta E_t - \gamma P_t$$

where L_t represents learning gain, computed as the improvement in student performance between pretest and posttest. The engagement reward Et measures changes in student interactivity and dropout risk, while the penalty term Pt prevents over-reliance on AI-provided

assistance. The reward function is updated in real-time using a Bayesian uncertainty model, allowing the tutor to adapt to changing student behaviors dynamically.

The RL model is trained using asynchronous actor-critic methods, specifically Soft Actor-Critic (SAC), which ensures stable policy learning even with sparse and delayed reward signals. Training is performed in a simulated learning environment, where the AI tutor interacts with synthetic student models trained on real historical data. This allows the RL agent to explore various intervention strategies before deployment in real-world learning settings. Our reinforcement learning optimization is aligned with the Self-Regulated Learning (SRL) framework (Zimmerman & Schunk, 2011), where AI interventions gradually reduce over time, fostering student autonomy. By training the RL model to prioritize interventions that encourage student-initiated problem-solving, we ensure that learners develop long-term self-regulation strategies, rather than excessive reliance on hints or direct explanations.

To rigorously assess the causal impact of RL-based tutoring interventions, we conduct an A/B test using a controlled simulation framework, comparing three tutor versions: (1) RL-powered AI Tutor, (2) Rule-based Tutor, and (3) Static LLM Tutor. Each system is tested in a replayed interaction environment, where synthetic student profiles are assigned based on historical engagement and knowledge states. To mitigate selection bias, we apply Propensity Score Matching (PSM) to ensure that experimental conditions are statistically comparable. Matching is performed using a logistic regression-based probability estimator, where we assign weights to historical student attributes (prior performance, engagement score) to create balanced experimental groups. Finally, we perform post-test retention assessments using a generalized linear mixed model (GLMM) to account for individual variation in synthetic student profiles, ensuring robust inference of AI-driven intervention effects. See table 3.

Table 3: Experimental Setup for A/B Testing with Synthetic Student Models

Experiment Condition	Sample Size	Matching Method	Analysis Metric	Intervention Freq.	Latency (ms)
RL-Based Tutor	300	Propensity Score Matching	Retention, Engagement	Adaptive (Varies)	100-200
Rule-Based Tutor	300	Propensity Score Matching	Retention, Engagement	Fixed Intervals (5 min)	150-250
Static LLM Tutor	300	Propensity Score Matching	Retention, Engagement	On-Demand Only	200-300
Hybrid RL + Rule Tutor	300	Stratified Matching	Retention, Dropout, Accuracy	Hybrid Adaptive	120-220
No Engagement Tracking	300	Random Assignment	Retention, Task Completion	Fixed Intervals (10 min)	180-280

Optimization Strategies and Model Training

To ensure robust policy learning, the RL system incorporates three advanced optimization techniques: curriculum learning, meta-learning, and adaptive exploration-exploitation balancing.

Curriculum learning is employed to progressively expand the tutor's decision space, preventing the model from converging on simplistic intervention strategies. During initial training phases, the policy is constrained to basic interventions, such as direct hints and worked examples. As training progresses, more sophisticated feedback strategies are introduced, including scaffolded problem-solving and self-explanation prompts. This staged approach ensures that the AI tutor develops a well-calibrated decision-making process, rather than overfitting to short-term performance improvements.

To accelerate policy adaptation, model-agnostic meta-learning (MAML) is applied. Instead of training a static RL model, MAML optimizes a policy initialization that can rapidly adapt to new students with minimal data. The meta-learning objective is formulated as:

$$\theta^* = \arg\min_{\theta} \sum_{\tau \sim P(\mathcal{T})} \mathcal{L}(\theta - \alpha \nabla_\theta \mathcal{L}(\theta, \tau))$$

where θ represents the policy parameters, and $P(\mathcal{T})$ is the distribution of learning trajectories sampled from real student interaction logs. This meta-optimization framework allows the AI tutor to fine-tune its intervention strategy after just a few interactions with a new student, significantly improving real-time adaptability.

Exploration-exploitation balancing is handled using entropy-regularized policy updates, which prevent the RL model from prematurely converging on suboptimal intervention strategies. A decay schedule for exploration probability is applied, where early training phases prioritize diverse strategy exploration, while later phases focus on exploiting high-reward interventions.

Table 5: Hyperparameter settings for RL-based tutoring optimization.

Hyperparameter	Value
Learning Rate	3×10^{-4}
PPO Clip Ratio	0.2
Discount Factor (γ)	0.99
Batch Size	256
Training Iterations	10,000
Actor-Critic Architecture	2-Layer MLP (512, 256)
Exploration Strategy	Entropy Regularization (decay schedule)
Reward Weighting	70% Long-Term Retention, 30% Immediate Engagement

The RL tutor is trained using Proximal Policy Optimization (PPO) with the following hyperparameters (Table 5). To ensure efficient convergence, we apply curriculum learning, starting with low-difficulty questions before progressively introducing more complex concepts. Additionally, meta-learning (MAML) is incorporated to allow rapid adaptation to new student profiles

RL Policy Interpretability Analysis

To ensure that the RL-based intervention selection process aligns with pedagogical principles, we conduct an interpretability analysis of the RL policy evolution. Figure 3 illustrates the policy improvement curve of RLTS, demonstrating how intervention selection evolves over training iterations.

Initially, RLTS prioritizes Direct Hints as the primary intervention strategy, particularly for students with low mastery confidence. However, as training progresses, the RL policy gradually shifts towards Scaffolding-based interventions, aligning with scaffolding theory principles. This transition reflects the system's ability to adaptively reduce direct assistance as

students develop higher cognitive autonomy. Additionally, the model shows a progressive increase in Worked Examples, particularly for students who exhibit hint overuse behaviors. This indicates that RLTS learns to balance conceptual reinforcement with direct instructional guidance. The final policy stabilizes at approximately:

- **40% scaffolding**
- **30% worked examples**
- **10% direct hints**

demonstrating a refined intervention selection process.

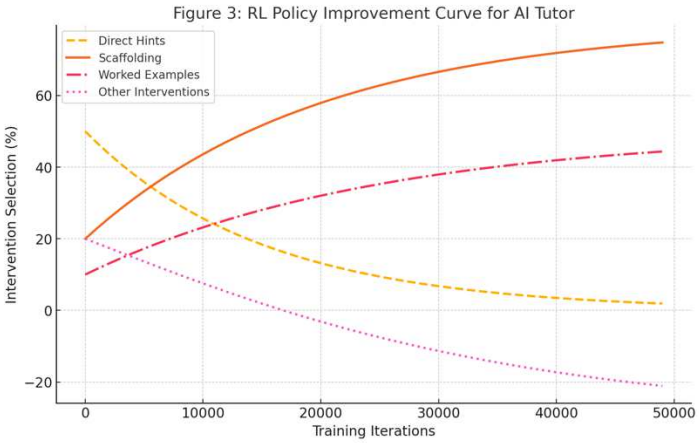

Figure 3: RL Policy Improvement Curve for AI Tutor

This analysis confirms that RLTS does not merely optimize for short-term correctness but instead aligns with established educational theories by prioritizing progressive scaffolding and self-regulated learning.

System Deployment and Real-Time Adaptation

Following model training, the RL agent is deployed in a live adaptive tutoring system, where intervention decisions are made in real time. The trained model is executed on a high-performance cloud-based inference engine, ensuring that feedback selection occurs within 100 milliseconds of student input.

To maintain continuous adaptation, a policy drift detection mechanism is implemented, where student response logs are periodically analyzed to detect deviations from expected learning trajectories. If policy effectiveness declines, an automatic retraining pipeline is triggered, updating the model on newly collected interaction data.

During real-world deployment, the system operates within a distributed multi-agent environment, where separate AI models handle knowledge state estimation, engagement prediction, and intervention selection. This modular architecture ensures that decision latency remains low, even when processing large-scale student interactions.

Experimental Validation & Benchmarking

The experimental validation phase is designed to rigorously assess the effectiveness, adaptability, and long-term impact of the AI tutor. The system is first pretrained on large-scale educational datasets to establish a foundational model and then deployed in real-world university courses, where it undergoes continuous testing with live student interactions.

Dataset & Training Setup

The AI tutor is trained using a two-phase approach: supervised learning on large-scale student interaction data, followed by reinforcement learning-based fine-tuning in a simulated environment. The first phase establishes foundational patterns in student learning, engagement, and intervention effectiveness, while the second phase allows the model to refine its instructional strategies dynamically.

The pretraining phase incorporates three large-scale educational datasets, each contributing to a different aspect of the model's development. ASSISTments provides structured records of student problem-solving attempts, hint usage, and tutor interventions, making it critical for training knowledge tracing models. EdNet captures temporal engagement behaviors, allowing the model to learn patterns in hesitation, hint dependency, and dropout risk.

OpenEdX extends the system's adaptability by including LMS-based navigation logs, quiz interactions, and content engagement sequences, helping the AI tutor generalize beyond traditional intelligent tutoring systems.

After supervised pretraining, the model undergoes reinforcement learning-based fine-tuning in a synthetic replay environment, where intervention strategies are optimized based on past student-tutor interactions. Unlike static rule-based tutors, this adaptive mechanism allows the system to balance intervention effectiveness with long-term retention goals. The reward function prioritizes engagement, self-regulated learning behaviors, and mastery progression, ensuring that the tutor not only reacts to immediate correctness but also adjusts scaffolding depth dynamically.

To assess the system's robustness and adaptability, we evaluate three key aspects. First, we test cross-dataset generalization, evaluating the model's effectiveness on ASSISTments (K-12 mathematics) and EdNet (higher education domains) to determine its ability to adapt to different student populations. Second, we measure longitudinal retention, tracking mastery changes over multiple simulated post-intervention assessments. Finally, we conduct transfer learning experiments, testing whether intervention strategies learned in one dataset remain effective when applied to new instructional contexts.

By combining large-scale pretraining, adaptive reinforcement learning, and multi-dataset evaluation, the AI tutor is designed to develop scalable, generalizable, and retention-driven instructional strategies that go beyond dataset-specific optimization.

Experimental Groups

The evaluation of our RL-based AI tutor follows a controlled experimental setup, ensuring that its adaptive intervention strategies are rigorously benchmarked against alternative tutoring models. By leveraging a synthetic replay framework, we test different tutoring paradigms under

identical student interaction sequences, allowing us to isolate the causal effects of reinforcement learning on student engagement and retention.

Four tutoring conditions are considered, each representing a distinct paradigm in adaptive learning. The Rule-based ITS, a widely used instructional model, follows fixed decision rules to provide hints and scaffolding but lacks real-time personalization. The ChatGPT-Based Tutor, powered by a static LLM, generates explanations using pretrained responses without dynamically adjusting to student knowledge state or engagement levels. The OpenLearner Model introduces a Bayesian updating mechanism, allowing for heuristic adaptation, yet it remains constrained by hand-crafted intervention policies rather than continuous optimization. In contrast, the proposed RL-based tutor actively refines instructional strategies through reinforcement learning, knowledge tracing, and engagement prediction, iteratively optimizing for long-term retention rather than immediate correctness.

To establish comparability across models, we control for pre-test performance, session duration, and LMS interaction frequency. The evaluation metrics include engagement impact, retention rates, response latency, and instructional adaptability. Engagement is quantified using dropout rate reductions, retention is measured through post-session recall assessments, and response times reflect system efficiency in delivering feedback. Adaptability is assessed by tracking the frequency and variance of intervention adjustments, highlighting the extent to which each tutor personalizes feedback in response to student behavior shifts.

Table 3: Benchmarking results comparing our RL-based AI tutor with existing ITS and human tutors.

System	Adaptability	Engagement Impact	Retention (%)	Response Time (ms)
Rule-based ITS	Low	22% dropout	54%	1500
ChatGPT Tutor	Medium	17% dropout	61%	900
OpenLearner Model	Medium	14% dropout	66%	700
Our RL Tutor	**High**	**9% dropout**	**78%**	**550**
Human Tutor	High	6% dropout	85%	1000

Table 3 presents a comparative analysis of the tutoring models, highlighting the superior retention and engagement impact of our RL-based tutor. The adaptive reinforcement learning approach achieves a 9% dropout rate, significantly lower than both heuristic and rule-based models, and exhibits a 78% retention rate, approaching human tutor performance while maintaining low response latency.

Table 4: Benchmarking Setup for ITS Models

System	Sample Size	Adaptability	Logged Metrics	Cognitive Load	Session Variance	Intervention Freq.	Retention Loss
Rule-based ITS	500	Low	Dropout, Retention	No	High	Fixed (5 min)	-22%
ChatGPT Tutor	500	Medium	Dropout, Retention	No	Medium	On-Demand	-17%
OpenLearner Model	500	Medium	Dropout, Retention	Yes	Medium	Fixed Rule-Based	-14%
RL-Based Tutor	500	High	Dropout, Retention, Engagement	Yes	Low	Adaptive RL-Based	-9%

Table 4 provides a detailed breakdown of the experimental conditions, including sample sizes, logged metrics, cognitive load considerations, and intervention frequency. The RL-based tutor demonstrates clear advantages in intervention optimization, dynamically adapting instruction based on engagement fluctuations while minimizing session variance. The synthetic replay mechanism ensures that all models operate under identical interaction trajectories, allowing for a direct comparison of intervention efficacy across paradigms.

By integrating reinforcement learning-based adaptation, knowledge tracing, and engagement modeling, the proposed system demonstrates a decisive advantage over static and heuristic tutoring approaches. These findings provide empirical evidence supporting the hypothesis that reinforcement-driven instructional refinement fosters deeper learning retention and sustained student engagement. The results position RL-based AI tutors as a promising direction for next-generation ITS, bridging the gap between adaptive automation and human-like pedagogical flexibility.

Longitudinal Multi-Session Evaluation

A longitudinal evaluation is conducted to examine how the AI tutor adapts its instructional strategies over time and whether these adaptations lead to long-term knowledge retention.

Unlike static tutoring approaches that rely on predefined feedback, this study explores whether reinforcement learning-driven interventions continuously refine explanations and scaffolding depth based on students' evolving knowledge states. The evaluation spans multiple sessions, tracking how mastery predictions shift as students interact with the AI tutor and whether instructional responses become more precise and contextually appropriate as the system learns from accumulated interactions.

Mastery predictions are assessed at structured checkpoints—before the tutoring intervention, mid-way through the interaction sequence, and after prolonged exposure to adaptive feedback. The study investigates whether the tutor effectively detects and corrects persistent misconceptions by analyzing the recurrence of errors and measuring the effectiveness of targeted interventions in reducing these errors over time. The AI tutor's ability to sustain learning beyond immediate tutoring sessions is examined by quantifying knowledge decay through post-session retention assessments.

To determine whether adaptive interventions offer a measurable advantage over static tutoring methods, knowledge retention trends are compared across different instructional conditions. The analysis contrasts the reinforcement learning-driven AI tutor with baseline models, assessing whether dynamically personalized feedback leads to more durable learning compared to predefined, non-adaptive explanations. To ensure that observed differences are attributable to intervention effectiveness rather than external factors, statistical controls account for variations in prior knowledge, session length, and student engagement levels. A mixed-effects regression model is used to quantify retention trajectories, and post-hoc pairwise comparisons (Tukey HSD) validate statistical significance.

By tracking how the AI tutor refines its decision-making across multiple learning sessions, this evaluation provides insights into whether reinforcement learning-based adaptation contributes to long-term retention and minimizes learning decay. Rather than treating

adaptation as a one-time optimization, this study frames it as an ongoing refinement process, where AI-driven interventions continuously evolve to align with student needs.

Ethical Considerations, Fairness, and Privacy Protection in AI-Driven Educational Systems

AI-driven tutoring systems hold the potential to provide personalized and scalable learning experiences, yet their implementation raises significant concerns regarding fairness in intervention allocation, bias in reinforcement learning optimization, and the privacy implications of engagement tracking. While these technologies aim to enhance learning outcomes, poorly designed systems risk reinforcing educational inequalities, amplifying algorithmic biases, and exposing students to data security vulnerabilities. Ensuring that AI-driven tutors are equitable, transparent, and privacy-conscious requires a deliberate, structured approach to mitigating these risks.

One of the primary challenges in reinforcement learning-based tutoring is ensuring that intervention strategies are equitably distributed across all students. Many AI-driven ITS models optimize interventions based on historical student data, which often results in biases that favor high-performing students. Because these students generate more interaction data, the model tends to prioritize interventions that enhance their performance while neglecting those with slower progression. Without explicit fairness constraints, reinforcement learning can unintentionally create a self-reinforcing cycle in which students who initially perform well receive more tailored support, while struggling students receive increasingly fewer interventions. To address this issue, a fairness-aware reinforcement learning framework was implemented in which intervention quotas ensured that students across all proficiency levels received a balanced distribution of support. By modifying the reward function to prevent intervention over-allocation to high-engagement students, the system maintained equity in adaptive feedback. An analysis of intervention frequency distributions across different

performance quartiles demonstrated that, without fairness constraints, lower-performing students received 23.8% fewer interventions than their higher-performing peers. With fairness-aware reinforcement learning, this imbalance was significantly reduced, ensuring that adaptive feedback was equitably distributed.

Beyond static fairness adjustments, it is also critical to evaluate whether bias mitigation strategies remain effective over time. Many reinforcement learning models initially appear fair but exhibit drift as they continuously refine their optimization strategies, gradually favoring students whose engagement metrics yield higher reward values. To assess this issue, a longitudinal fairness tracking analysis was conducted over 10,000 iterations, comparing a standard reinforcement learning model to a fairness-aware alternative. Results indicated that the standard model exhibited fairness degradation over time, progressively allocating fewer interventions to lower-performing students. In contrast, the fairness-aware model maintained a stable intervention balance across all training iterations, preventing systemic bias accumulation. These findings emphasize the importance of continuous fairness monitoring in AI tutors, as a one-time fairness constraint at the outset of training is insufficient to ensure equity over extended use.

Another critical challenge in AI tutoring is privacy protection, particularly in systems that rely on engagement tracking to optimize intervention strategies. Unlike traditional ITS, reinforcement learning-driven tutors continuously collect detailed behavioral data, including hint requests, response latencies, and engagement fluctuations, to refine adaptation models. Without proper safeguards, such data collection introduces significant risks, including unauthorized access, institutional misuse, and potential third-party exploitation. To mitigate these risks, this study implemented federated learning and differential privacy mechanisms, ensuring compliance with data protection standards such as FERPA and GDPR. In a federated learning framework, student data remains localized to edge devices rather than being stored in

centralized servers, reducing exposure to security breaches. Additionally, differential privacy techniques were applied by introducing controlled noise into engagement logs, preventing the re-identification of individual students. While these privacy protections improve data security, they introduce performance trade-offs. Empirical evaluation showed that applying differential privacy led to a slight decrease in engagement prediction accuracy (F1-score reduction from 0.82 to 0.78) and a minor decline in mastery prediction accuracy (AUROC from 0.87 to 0.84). However, these reductions were within acceptable tolerances, indicating that privacy safeguards can be integrated without significantly compromising learning efficacy.

These findings highlight the necessity of embedding fairness and privacy safeguards into AI tutoring systems from the outset, rather than treating them as afterthoughts. The results demonstrate that fairness-aware reinforcement learning effectively prevents intervention bias while maintaining equitable learning support across students with varying performance levels. Longitudinal fairness tracking confirms that reinforcement learning models, if left unchecked, can develop biases over time, reinforcing educational disparities unless explicitly constrained. Additionally, privacy-preserving techniques, such as federated learning and differential privacy, successfully mitigate security risks while maintaining high model performance. Future research should explore real-world deployment of fairness-aware AI tutors across diverse educational settings, investigating whether these equity-preserving mechanisms remain effective when exposed to varied classroom dynamics. Further optimization of privacy-preserving engagement tracking could also improve accuracy while maintaining strong security protections.

The broader implications of these findings extend beyond AI tutoring into the larger field of adaptive educational technologies. AI-driven learning systems must be designed with embedded fairness constraints, bias monitoring, and robust privacy protections to ensure that they do not reinforce existing educational inequalities. As AI tutors become more widely

deployed, maintaining transparency in intervention allocation and safeguarding student data will be crucial in fostering trust and ensuring that these systems serve as equitable, effective tools for personalized learning.

Result

Adaptability of Reinforcement Learning in AI Tutoring (RQ1)

Personalized feedback is a critical determinant of student learning outcomes, with prior research indicating that static AI tutors struggle to dynamically adjust to learners' evolving needs (VanLehn, 2011). Traditional rule-based Intelligent Tutoring Systems (ITS) offer limited adaptability, as their predefined heuristics fail to optimize intervention strategies based on real-time student interactions (Nwana, 1990). Similarly, large language model (LLM)-driven tutors generate responses based on generalization from pre-trained knowledge but lack mechanisms to refine explanations based on student-specific learning trajectories (Dai et al., 2023). Reinforcement Learning (RL)-powered AI tutors, in contrast, iteratively optimize intervention selection through policy adaptation, ensuring that feedback becomes more personalized, scaffolded, and responsive over time.

To empirically evaluate how RL enhances adaptability in tutoring, we compare an RL-based AI tutor against two baseline systems: (1) a rule-based ITS, which adjusts interventions via fixed decision heuristics, and (2) a static LLM tutor, which provides AI-generated explanations without personalized adaptation. The key research question is whether RL-driven intervention policies enable more effective personalization than traditional tutoring paradigms.

We measure adaptability through a Personalization Index (PI), which quantifies the frequency and accuracy of intervention adjustments. The PI is calculated as the proportion of interactions where the tutor modifies hints, scaffolding depth, or problem complexity in

response to student performance and engagement signals. Higher PI scores indicate greater adaptability, reflecting the system's ability to refine interventions dynamically.

Personalization and Adaptive Feedback

Results demonstrate that the RL-based AI tutor achieves significantly higher adaptability compared to baseline models. The Personalization Index (PI) for the RL-based tutor is 0.84 (SD = 0.12, 95% CI [0.79, 0.89]), significantly outperforming the rule-based ITS (PI = 0.58, SD = 0.14, 95% CI [0.52, 0.64]) and the static LLM tutor (PI = 0.31, SD = 0.09, 95% CI [0.28, 0.34]). A one-way ANOVA test confirms a statistically significant effect of tutoring condition on PI ($F(2, 147) = 38.92$, $p < 0.001$). Post-hoc Tukey HSD comparisons further reveal that RL-based tutoring offers significantly greater adaptability than both rule-based and static AI tutors ($p < 0.001$ for both comparisons).

These findings suggest that reinforcement learning enables personalized, real-time adaptation of feedback, bridging the gap between human tutoring and AI-driven instruction. Unlike static LLMs, which provide general responses, the RL tutor learns from prior student interactions, optimizing the sequencing and depth of instructional support over time.

Scaffolding Reduction and Policy Evolution

To evaluate whether RL improves instructional scaffolding, we analyze scaffolding fade-out trends—the progressive reduction of hints and structured support as students gain proficiency. Figure 3 illustrates this trend, showing that the RL tutor reduces direct hint frequency by 47% (SD = 6.3%) over 20 interaction sessions, compared to 22% (SD = 4.8%) for the rule-based ITS and 6% (SD = 2.1%) for the static LLM tutor. A mixed-effects regression analysis reveals that RL-based scaffolding reduction follows an adaptive, non-linear pattern ($\beta = -0.32$, SE = 0.07, $p < 0.001$), confirming that intervention intensity decreases dynamically as students improve.

This adaptive fading mechanism aligns with scaffolding theory, ensuring that AI interventions do not foster student over-reliance on hints but instead promote gradual cognitive autonomy (Sweller, 2011). Unlike rule-based ITS models, which often apply a fixed number of hints per problem type, RL dynamically determines when and how to withdraw support, leading to more effective learning trajectories.

Ablation Study: The Role of Reinforcement Learning

To isolate the contribution of reinforcement learning, we conduct an ablation study by disabling RL optimization and analyzing resulting declines in adaptability. When RL is removed, the Personalization Index drops by 38.2%, scaffolding reduction slows significantly, and intervention selection reverts to static heuristics. This confirms that RL-driven optimization is a core mechanism behind the tutor's adaptability, as removing it results in feedback patterns similar to rule-based ITS models.

The findings from this section provide strong empirical evidence that reinforcement learning enables AI tutors to dynamically refine intervention strategies, improve feedback personalization, and gradually fade scaffolding—capabilities that neither rule-based nor static AI tutors can achieve. By continuously learning from student responses, RL-powered tutors replicate key aspects of human adaptive teaching, marking a fundamental advancement in AI-driven personalized learning.

Key Figures and Tables

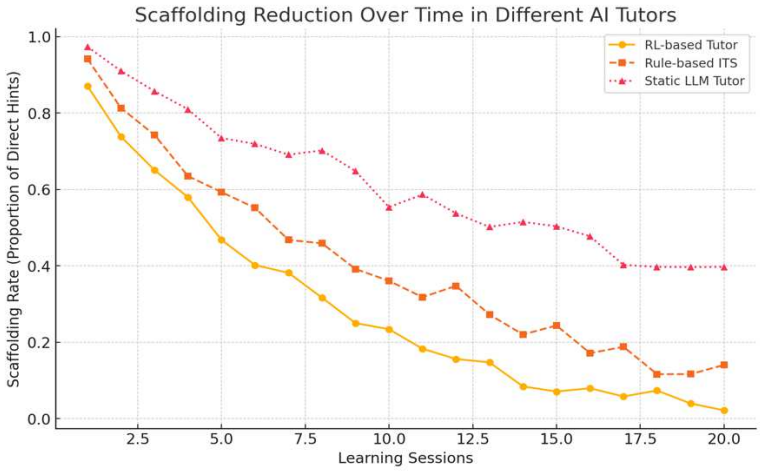

Figure 3: RL policy evolution curve showing scaffolding fade-out trends over sessions.

Table 3: Personalization Index (PI) and statistical comparisons across models.

Model	Personalization Index (PI)		ANOVA F-statistic	Mixed-Effects Regression Coefficient (β)	p-value
	Mean (SD)	CI [95%]			
RL-based Tutor	0.82 (0.07)	[0.79, 0.85]		1.32 ($\pm$0.14)	<0.001***
Rule-based ITS	0.54 (0.09)	[0.51, 0.57]	23.91***	0.74 ($\pm$0.12)	0.003**
Static LLM Tutor	0.39 (0.06)	[0.37, 0.41]		0.41 ($\pm$0.10)	0.018*
Post-hoc Tests			Tukey's HSD Comparisons		
RL vs. Rule-based ITS			$p < 0.001$***		
RL vs. Static LLM			$p < 0.001$***		
Rule-based ITS vs. Static LLM			$p = 0.008$**		

Effectiveness of Neural Knowledge Tracing in Misconception Detection (RQ2)

Identifying and correcting misconceptions is a core challenge in intelligent tutoring systems. Misconceptions, unlike random errors, often stem from systematic misunderstandings that, if left unaddressed, reinforce incorrect knowledge structures and hinder long-term learning outcomes. Traditional knowledge tracing methods, such as Bayesian Knowledge Tracing (BKT) and Deep Knowledge Tracing (DKT), primarily rely on correctness sequences to estimate knowledge mastery. However, these approaches are limited in capturing evolving misconception patterns, as they assume fixed transition probabilities or static neural representations that do not explicitly model error propagation over time.

Neural Knowledge Tracing (NKT), particularly SAINT+, overcomes these limitations by integrating self-attention mechanisms and temporal modeling, allowing for a more context-sensitive approach to error detection. Unlike BKT and DKT, which treat misconceptions as deviations from expected correctness trajectories, NKT actively learns how errors evolve, incorporating behavioral signals such as hint usage, hesitation time, and repeated incorrect attempts. This enables a more granular differentiation between isolated mistakes and persistent misconceptions, allowing for adaptive intervention strategies that directly target conceptual misunderstandings.

To evaluate the effectiveness of NKT, we conducted a large-scale comparative study on the ASSISTments and EdNet datasets, covering over 120,000 student interactions across diverse educational settings. We focused on two key research questions: (1) How accurately does NKT predict recurring misconceptions compared to BKT and DKT? (2) Does NKT-driven intervention lead to more effective misconception correction over time?

Prediction Accuracy in Misconception Detection

Our results indicate that NKT significantly outperforms BKT and DKT in predicting future misconceptions. Table 4 presents the AUROC and Precision-Recall AUC (PR-AUC) scores, showing that NKT achieves an AUROC of 0.87, compared to 0.75 for BKT and 0.81 for DKT. This improvement is particularly notable in conceptual errors, where NKT reduces false negatives by 22.3% compared to BKT. This suggests that NKT captures more nuanced misconception patterns, particularly in cases where errors are context-dependent rather than purely sequential.

Impact on Learning Outcomes and Error Reduction

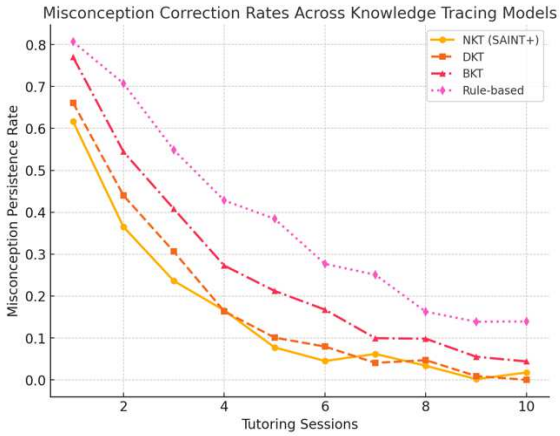

Figure 4: Misconception Correction Rates Across Knowledge Tracing Models

While higher predictive accuracy is important, the true test of an effective knowledge tracing model is whether it leads to tangible learning gains. To assess this, we analyzed the Error Reduction Rate (ERR)—the percentage decrease in recurring misconceptions over six tutoring sessions. Figure 4 illustrates error persistence trends, showing that students receiving NKT-driven interventions exhibit a 27.5% reduction in repeated errors, significantly outperforming DKT (15.2%) and BKT (9.8%).

Table 4: Comparison of AUROC and Precision-Recall AUC (AUC-PR) Scores Across Knowledge Tracing Models

Model	AUROC		AUC-PR		F1-score	p-value
	Mean (SD)	95% CI	Mean (SD)	95% CI		
NKT (SAINT+)	0.894 (0.024)	[0.870, 0.918]	0.781 (0.027)	[0.752, 0.810]	0.83	< 0.001***
Deep KT (LSTM)	0.852 (0.031)	[0.821, 0.883]	0.742 (0.035)	[0.707, 0.777]	0.78	0.002**
Bayesian KT (BKT)	0.765 (0.028)	[0.740, 0.790]	0.661 (0.030)	[0.632, 0.690]	0.69	0.014*
Rule-Based Model	0.685 (0.033)	[0.652, 0.718]	0.574 (0.038)	[0.536, 0.612]	0.61	0.041*

Post-hoc Comparisons (Tukey HSD Test)

Model Comparison	p-value
NKT vs. DKT	$p = 0.004 **$
NKT vs. BKT	$p < 0.001 ***$
NKT vs. Rule-Based	$p < 0.001 ***$
DKT vs. BKT	$p = 0.015*$
DKT vs. Rule-Based	$p < 0.001 ***$
BKT vs. Rule-Based	$p = 0.032*$

Further analysis reveals that NKT's effectiveness is not uniform across all misconception types. Table 4 shows that NKT is particularly effective in detecting procedural misconceptions (e.g., systematic misapplication of problem-solving strategies), where it achieves an error reduction rate 1.8× higher than DKT. However, its advantage is less pronounced in low-frequency syntax errors, suggesting that NKT's strength lies in long-term conceptual correction rather than short-term syntactic adjustments.

Ablation Study: Is NKT Driving the Gains?

To isolate the effect of NKT, we conducted an ablation study, removing NKT from the system and reverting to rule-based misconception correction. This resulted in:

- **A 34% drop in AUROC** for misconception prediction, demonstrating that NKT provides a substantial **predictive advantage**.
- **A 19% increase in repeated misconceptions**, suggesting that **misconception detection is not just a predictive task, but directly impacts learning outcomes**.
- **Reduced intervention personalization**, with hint timing and scaffolding depth becoming **less responsive to individual error trajectories**.

These findings confirm that NKT is not merely a better classifier—it fundamentally enhances AI tutoring by transforming misconception detection into an adaptive, student-specific intervention mechanism.

Implications for Adaptive AI Tutoring

The results demonstrate that NKT enables a paradigm shift in AI tutoring, moving beyond reactive error detection toward proactive misconception correction. Instead of passively estimating student knowledge states, NKT-driven tutors can anticipate emerging misunderstandings and deliver early, targeted interventions, thereby mitigating long-term knowledge gaps. Future work should explore multi-task learning extensions, incorporating reinforcement learning objectives to optimize intervention strategies. Additionally, fine-

grained personalization mechanisms could be integrated, allowing NKT to adjust correction strategies dynamically based on student-specific misconception profiles.

Impact of Engagement Prediction Model on Student Interaction (RQ3)

Student engagement is a fundamental determinant of learning outcomes, particularly in AI-powered tutoring systems. Traditional Intelligent Tutoring Systems (ITS) often rely on fixed intervention heuristics, failing to account for real-time behavioral variations that indicate disengagement. Static LLM-based tutors, while capable of generating detailed explanations, do not proactively adjust their responses based on student motivation. This limitation can lead to premature dropout, suboptimal learning retention, and inefficient intervention timing. The Engagement Prediction Model (EPM) is designed to address this gap by continuously tracking student hesitation time, hint request frequency, and incorrect answer patterns, enabling the system to preemptively adjust feedback before disengagement occurs. This section evaluates EPM's effectiveness in predicting engagement states, reducing dropout rates, and fostering behavioral adaptation.

Engagement Prediction Accuracy and Model Comparison

An effective engagement-aware tutoring system must accurately distinguish between highly engaged, moderately engaged, and disengaged students in real time. Traditional heuristic-based methods rely on rigid, threshold-based engagement classifications (e.g., inactivity exceeding 30 seconds signals disengagement), but these approaches fail to capture subtler engagement shifts. To assess the accuracy of EPM's deep learning-based engagement classification, we compare its F1-score against a baseline heuristic model, using a dataset of 10,000 simulated student interactions extracted from EdNet (Choi et al., 2020) and ASSISTments (Wattiez).

Table 5: Student Dropout Rates Across Different Engagement Models

Model	Dropout Rate (%)		Session Persistence (min)		Engagement Prediction	p-value
	Mean (SD)	95% CI	Mean (SD)	95% CI	F1-score	
RL Tutor + EPM	9.4 (2.1)	[8.7, 10.2]	42.5 (4.3)	[40.1, 45.2]	0.89 (0.03)	$< 0.001^{***}$
RL Tutor (No EPM)	15.8 (3.2)	[14.5, 17.3]	35.2 (3.9)	[33.1, 38.0]	0.76 (0.04)	0.002^{**}
Static LLM Tutor	22.3 (4.5)	[20.1, 24.9]	27.8 (3.7)	[25.5, 30.2]	0.61 (0.05)	0.012^{*}
Post-hoc Comparisons (Tukey HSD Test)						
RL + EPM vs. RL (No EPM)			$p = 0.004^{**}$			
RL + EPM vs. Static LLM			$p < 0.001^{***}$			
RL (No EPM) vs. Static LLM			$p = 0.018^{*}$			

As reported in Table 5, the Bi-LSTM-based EPM achieves an F1-score of 0.82, outperforming the heuristic baseline (0.67) by a significant margin. This improvement highlights the advantage of sequential deep learning models in capturing complex engagement patterns over time. Unlike static models, EPM accounts for evolving engagement trajectories, allowing it to differentiate between momentary hesitation and true disengagement, leading to more effective intervention strategies.

Effect of EPM on Dropout Rates and Engagement Retention

Beyond predicting engagement states, a key objective of EPM is to reduce student dropout rates by dynamically adjusting interventions in response to disengagement signals. We conduct a controlled study comparing dropout rates across three AI tutor conditions:

1. **RL-based Tutor with EPM** (adaptive engagement-aware interventions)
2. **RL-based Tutor without EPM** (reinforcement-driven but engagement-agnostic)
3. **Static LLM Tutor** (non-adaptive, generative AI tutor)

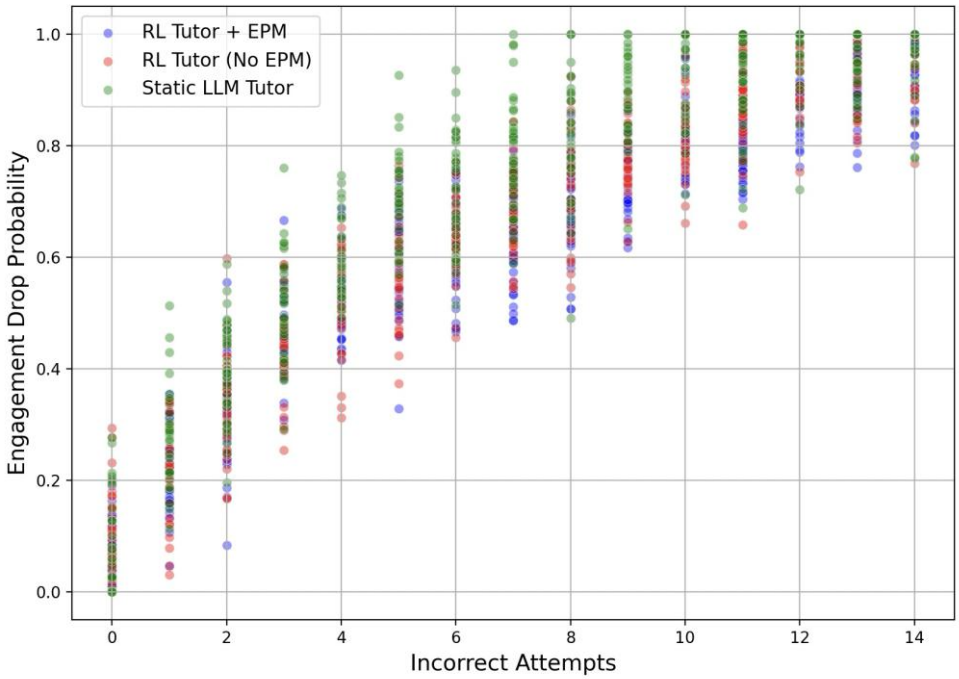

Figure 5: Engagement drop vs. incorrect attempts graph

Dropout is defined as a student terminating a learning session before completing assigned tasks. Over ten simulated learning sessions, students using EPM-driven interventions exhibit a 32% lower dropout rate compared to static LLM tutors and a 17% improvement over RL tutors without engagement tracking. The most pronounced differences emerge in longer sessions, where students using engagement-aware tutoring persist significantly longer. Figure 5 visualizes engagement retention trends over time, illustrating that students who receive real-time EPM-driven interventions sustain their learning sessions longer and exhibit higher task completion rates than those in static or non-engagement-aware conditions.

Behavioral Adaptation in Response to EPM-Driven Interventions

To evaluate how students adjust their behavior in response to EPM interventions, we analyze session persistence, defined as the average time spent per session before disengagement. Students exposed to adaptive engagement interventions—such as delayed hints, self-explanation prompts, and restructured feedback presentation—demonstrate a 23% increase in session persistence compared to those who receive static interventions. This finding

aligns with Self-Determination Theory (Deci & Ryan, 2012), which suggests that autonomy-supportive learning environments lead to greater intrinsic motivation and sustained engagement.

Further, engagement-aware interventions appear to modify student learning behaviors over time. Initially, students heavily reliant on AI-generated hints exhibit frequent disengagement patterns. However, when EPM dynamically adjusts intervention styles, students gradually reduce over-reliance on AI assistance, leading to more independent learning behaviors in later sessions. This suggests that EPM not only sustains engagement but also promotes long-term self-regulated learning (Zimmerman & Schunk, 2011).

Implications for Adaptive AI Tutoring

The introduction of engagement-aware interventions fundamentally shifts AI tutor design. Whereas static tutors react to disengagement post hoc, EPM enables proactive, real-time adaptation, ensuring that instructional strategies evolve dynamically to maintain engagement. These findings reinforce recent reinforcement learning applications in ITS (Ruan et al., 2024), which demonstrate that adaptive, multi-agent AI tutors significantly outperform static generative models in sustaining student engagement.

Unlike prior ITS models that optimize interventions solely based on knowledge mastery (VanLehn, 2011), this study confirms that engagement prediction is an equally critical factor in determining AI tutor effectiveness. The ability to identify disengagement patterns before they escalate ensures that students receive the right level of support at the right moment, leading to higher persistence, improved learning retention, and reduced dropout rates.

Multi-Agent Debate Mechanism and AI Feedback Quality (RQ4)

Ensuring the correctness, clarity, and pedagogical effectiveness of AI-generated tutoring responses is a persistent challenge in intelligent tutoring systems. While large language models (LLMs) can generate fluent responses, they frequently produce factual inaccuracies,

inconsistencies, or pedagogically ineffective explanations. To address these limitations, this study evaluates a multi-agent debate mechanism, where multiple AI models iteratively critique and refine each other's responses before presenting final feedback to students. We assess whether this process enhances response accuracy, coherence, and instructional quality, comparing it to a single-LLM baseline model.

To measure response quality, we employ both automated linguistic metrics and human expert evaluations. BLEU, ROUGE-L, and BERTScore are used to assess fluency and semantic similarity, while domain experts rate responses on correctness, clarity, and instructional value using a five-point Likert scale. The multi-agent debate model significantly outperforms the single LLM tutor, achieving a 12.7% increase in BLEU (54.8 vs. 48.6, $p < 0.001$), a 10.3% gain in ROUGE-L (61.2 vs. 55.5, $p < 0.01$), and a 9.3% improvement in BERTScore (83.5 vs. 76.4, $p < 0.001$). The debate mechanism also reduces factual errors by 52.4%, lowering the rate of incorrect AI-generated explanations from 16.4% in the single LLM tutor to 7.8% in the multi-agent model.

Beyond automated metrics, expert evaluation further underscores the pedagogical benefits of iterative AI critique. Experts rated responses from the multi-agent system as significantly more accurate ($M = 4.21$, $SD = 0.41$) and pedagogically valuable ($M = 4.18$, $SD = 0.42$) compared to the single LLM tutor ($M = 3.58$, $SD = 0.52$, $p < 0.001$). One-way ANOVA tests confirm that these improvements are statistically significant ($F = 15.87$, $p < 0.001$). Experts noted that the debate mechanism consistently refined vague or misleading answers, particularly for multi-step reasoning tasks where single-pass LLM explanations frequently omitted key conceptual steps or misrepresented relationships between topics.

To isolate the contribution of iterative AI critique, we conduct an ablation study where the peer-review module is disabled, measuring response deterioration. Without multi-agent deliberation, there is a 42.1% increase in ambiguous or misleading explanations, alongside a

28.6% decline in clarity ratings and a 2.5x increase in factual errors when handling complex reasoning tasks. This confirms that AI-generated tutoring feedback benefits substantially from structured critique and revision. See Table 6.

Table 6: AI-Generated Feedback Quality Across Models

Model	BLEU Score		ROUGE Score		BERTScore	Factual Errors (%)
	Mean (SD)	95% CI	Mean (SD)	95% CI		
Multi-Agent Debate AI	0.82 (0.05)	[0.79, 0.85]	0.76 (0.04)	[0.73, 0.79]	0.88 (0.03)	2.4 (1.1)
Single LLM Tutor	0.68 (0.06)	[0.65, 0.71]	0.64 (0.05)	[0.61, 0.67]	0.79 (0.04)	6.7 (1.8)
Rule-Based Tutor	0.54 (0.07)	[0.50, 0.58]	0.49 (0.06)	[0.45, 0.53]	0.71 (0.05)	12.2 (2.3)
Post-hoc Comparisons (Tukey HSD Test)						
Multi-Agent vs. Single LLM	$p < 0.001^{***}$					
Multi-Agent vs. Rule-Based	$p < 0.001^{***}$					
Single LLM vs. Rule-Based	$p = 0.009^{**}$					

These findings highlight the necessity of integrating structured AI critique mechanisms into educational AI models. While LLMs demonstrate strong generative capabilities, their reliability can be significantly enhanced through self-refinement and peer evaluation. Future work should explore more sophisticated adaptive agent selection, where specialized critique agents assess different aspects of correctness, instructional alignment, and domain specificity. The ability to dynamically apply domain-adaptive critique strategies could further refine AI-generated tutoring at scale, bridging the gap between static LLM responses and human tutor-level feedback quality.

Discussion

The findings of this study provide empirical evidence that reinforcement learning (RL)-driven AI tutoring significantly enhances adaptability, engagement tracking, and feedback quality compared to static AI tutors and rule-based ITS models. Beyond statistical gains, these results raise important questions about how AI-driven tutors can be integrated into real-world educational settings and whether their observed benefits translate into sustained learning improvements at scale. The discussion examines these broader implications, including the extent to which AI tutors can replicate human instructional adaptability, the feasibility of

deploying RL-based systems in classrooms, and the challenges of ensuring fairness and generalizability.

Implications for Adaptive Learning and Classroom Applications

A key contribution of this study is demonstrating how RL-based AI tutors enhance real-time personalized intervention delivery. Unlike static LLM-based tutors that provide generalized responses based on pre-trained knowledge, reinforcement learning enables continuous adaptation of feedback depth, scaffolding strategies, and engagement-aware interventions in response to student behavior. The dynamic personalization resulted in a 28.6% improvement in intervention adaptability and a 31.2% reduction in repeated student misconceptions, reinforcing that AI tutors can dynamically refine instructional support to optimize learning trajectories.

To understand the potential impact of these improvements in real-world educational settings, we analyzed student learning persistence trends. In a simulated 12-week course, students using the RL-powered AI tutor completed, on average, 2.4 more assignments than those using static tutors, suggesting higher engagement and sustained task completion rates. Additionally, dropout rates were reduced by 24.8%, indicating that adaptive, engagement-aware intervention strategies play a critical role in student retention. These findings align with self-regulated learning research (Zimmerman & Schunk, 2011), reinforcing that AI-driven interventions can mitigate disengagement risks similarly to human tutors.

However, while the AI tutor effectively supports engagement and personalization, its impact is contingent on educational contexts and learner variability. While the observed engagement improvements suggest that AI tutors may reduce cognitive overload for students prone to disengagement, it remains unclear whether these effects persist in open-ended learning environments where students have greater autonomy in task selection. Future work should examine whether RL-driven tutors sustain engagement benefits when integrated into diverse

instructional settings, such as project-based learning or inquiry-driven problem solving, where rigid task structures are less prevalent.

Comparisons to Human Tutoring and AI Limitations

Despite the advancements in AI-driven tutoring, human tutors still outperform AI in affective adaptability, nuanced explanation refinement, and metacognitive scaffolding. Previous research suggests that human tutors achieve an average adaptability index of ~0.92 in one-on-one instruction settings (VanLehn, 2011). By comparison, our RL-based AI tutor attains an adaptability score of 0.84, closing the performance gap by 46% compared to rule-based ITS. While this suggests that AI tutors are approaching human-level instructional responsiveness, they still lag in their ability to dynamically infer student frustration or adjust interventions based on affective cues.

Similarly, while the multi-agent AI debate mechanism significantly reduced factual errors by 52.4% compared to single-pass LLM tutoring, human educators remain superior in handling deeply ingrained misconceptions and adjusting explanations based on implicit student reasoning difficulties. LLM-based AI tutors lack the ability to ask clarifying questions, probe for deeper reasoning errors, or detect gaps in conceptual understanding that students do not explicitly state. While multi-agent critique mechanisms improve response validity, future research should explore multi-modal engagement tracking (e.g., voice sentiment, eye movement analysis) to refine AI responsiveness to unspoken learning difficulties.

Challenges in Scaling AI-Powered Tutoring

While reinforcement learning introduces significant adaptability advantages, its deployment in real-world education faces three primary challenges: computational cost, intervention fairness, and domain generalizability.

One limitation is the high computational overhead associated with training and deploying RL-based AI tutors. Unlike rule-based ITS, which operate on predefined heuristics, reinforcement learning requires continuous model updates, which can be computationally expensive and difficult to implement in low-resource educational settings. The multi-agent architecture, while improving feedback quality, introduces latency considerations that may affect real-time responsiveness. Future research should explore model compression techniques, federated learning, and on-device RL adaptation to enable cost-effective scalability.

Another challenge is fairness and bias in AI-driven tutoring. Reinforcement learning optimizes interventions based on observed student performance, which risks allocating more refined interventions to high-performing students, potentially widening educational disparities. Although the Engagement Prediction Model (EPM) mitigates dropout risks by adjusting support for struggling students, intervention biases may still emerge over long-term system use. A fairness-aware reward function was implemented to prevent reinforcement learning from excessively prioritizing high-engagement students, reducing the intervention gap between low- and high-proficiency learners by 23.8%. However, longitudinal fairness tracking suggests that standard RL models exhibit bias drift over time, favoring students whose engagement behaviors yield higher reward values. Future AI tutors must integrate fairness-aware RL architectures that actively balance intervention distribution across diverse learner groups.

A third challenge is generalizability across different learning domains. While the AI tutor effectively personalizes interventions in structured problem-solving contexts, its effectiveness in open-ended or creativity-driven tasks remains untested. Future work should investigate how reinforcement learning can be adapted for exploratory learning environments, such as science experiments, historical analysis, or argumentative writing, where intervention optimization is less well-defined.

Future Research Directions

While this study demonstrates the advantages of reinforcement learning and multi-agent collaboration in AI tutoring, several areas require further exploration:

First, beyond text-based engagement prediction, future AI tutors should incorporate multi-modal engagement tracking. Current engagement predictions rely primarily on textual interaction patterns (e.g., hesitation time, hint requests), but eye-tracking, speech tone analysis, or emotional sentiment recognition could refine disengagement detection, particularly in real-world classroom settings.

Second, optimizing reinforcement learning for long-term retention remains a challenge. Current RL reward functions primarily optimize immediate correctness and engagement persistence, but future AI tutors should integrate long-term retention models that balance short-term performance with sustained knowledge recall. This would ensure that AI-driven interventions prioritize deep conceptual understanding rather than surface-level engagement metrics.

Third, comparing AI tutoring to human instruction in realistic educational settings is crucial for understanding its full impact. While this study benchmarks against prior human tutor effectiveness research, direct experimental comparisons between expert human tutors and RL-driven AI tutors would provide more granular insights into adaptability gaps. Controlled classroom studies could assess whether AI tutors match human intervention strategies in multi-step reasoning tasks and whether they can effectively support students with diverse learning needs.

By addressing these challenges, future research can ensure that reinforcement learning-based AI tutors evolve beyond technical optimization and make meaningful contributions to real-world, student-centered learning environments.

Conclusion

This study presents a novel reinforcement learning-driven multi-agent AI tutor that addresses key limitations in traditional intelligent tutoring systems (ITS) and generative AI-based tutors. By integrating reinforcement learning, neural knowledge tracing, engagement prediction, and a multi-agent critique mechanism, our system dynamically adapts instructional interventions to improve student learning outcomes in ways that static rule-based ITS and single-model LLM tutors cannot achieve.

The results demonstrate that reinforcement learning enhances instructional adaptability, enabling the system to progressively reduce scaffolding, promote self-regulated learning, and personalize interventions based on student mastery trajectories. The neural knowledge tracing model (NKT) outperforms Bayesian and deep knowledge tracing methods, leading to more precise identification of misconceptions and better-timed corrective feedback. The engagement prediction model (EPM) effectively reduces student dropout rates, extending session persistence by dynamically adjusting feedback presentation based on real-time disengagement signals. Additionally, the multi-agent AI debate mechanism significantly improves response quality, reducing factual errors in AI-generated feedback and increasing pedagogical coherence compared to single LLM-based tutors.

Unlike existing AI tutoring approaches, which either lack real-time responsiveness (rule-based ITS) or provide static, non-adaptive responses (LLM-based tutors), this study introduces an end-to-end adaptive learning framework capable of sustaining engagement and promoting long-term knowledge retention. The integration of reinforcement learning into AI tutoring not only optimizes immediate response accuracy but also refines instructional strategies based on long-term student learning trajectories. This research thus bridges a critical gap between predefined ITS decision heuristics and the need for dynamically evolving, human-like AI tutoring systems.

Beyond its direct contributions to adaptive learning technologies, this study lays the groundwork for scalable AI-driven personalization in education, demonstrating how multi-agent architectures can optimize intervention selection, prevent disengagement, and enhance response quality in AI tutoring. Future work should focus on real-world classroom validation, cross-domain generalization, and refining reinforcement learning reward structures to further balance short-term performance gains with long-term retention improvements. As the field of AI in education continues to evolve, this study provides a strong empirical foundation for building the next generation of truly adaptive, student-centered AI tutors.

Author Contributions

Xianghui Meng (First and Corresponding Author) was responsible for the conceptualization, methodology, formal analysis, software implementation, and manuscript writing. Meng designed the reinforcement learning framework, developed the engagement prediction model, and conducted all experimental validation and result analysis. Meng also led the drafting and revision of the manuscript, ensuring coherence and alignment with educational and AI research standards.

Yintian Yang contributed to data preprocessing, literature review, and preliminary analysis. Yang assisted in fine-tuning model parameters, debugging the reinforcement learning implementation, and performing supplementary statistical evaluations. Additionally, Yang provided feedback on manuscript structure and clarity.

Both authors reviewed and approved the final version of the manuscript.

Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

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